

UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the
Interstate Transmission System

Docket No. RM26-4-000

COMMENTS OF THE ADVANCED ENERGY MANAGEMENT ALLIANCE (AEMA)

I. INTRODUCTION

AEMA¹ respectfully submits these comments in response to the Notice Inviting Comments regarding the proposed Advance Notice of Proposed Rulemaking (ANOPR) regarding the timely and orderly interconnection of large loads to the interstate transmission system.²

These comments are submitted by the Advanced Energy Management Alliance (AEMA) and a coalition of companies and organizations dedicated to advancing economic growth through advanced energy solutions while improving the efficiency and reliability of the nation's electric system. Since its founding more than a decade ago, AEMA has focused on expanding access to demand response and distributed energy resources (DERs) to lower costs and improve grid stability. As the power system enters a new era of large-scale computing and manufacturing growth, AEMA's work now extends to integrating flexible industrial and data center loads that can enhance reliability while supporting economic expansion.

II. SUMMARY

AEMA strongly supports the Federal Energy Regulatory Commission's (Commission) initiative to establish standardized, transparent, and efficient procedures for interconnecting large loads to the transmission system. Rapid growth in artificial intelligence, semiconductor manufacturing, and advanced data-center infrastructure has created new patterns of electricity demand that require faster and more predictable interconnection processes.

AEMA agrees with the urgent need for reform articulated in the ANOPR. AEMA urges the Commission to create a Flexible Load Fast Track (FLFT) for facilities that can verifiably adjust their consumption in response to system conditions and harness a broad set of technologies to

¹ These comments are submitted jointly by the Advanced Energy Management Alliance (AEMA) and the following companies: Emerald AI, Voltus, Calibrant Energy, Fluence (collectively, the "DC Flexibility Joint Filers"). The DC Flexibility Joint Filers are in broad agreement on the principles and recommendations set forth herein, particularly the urgent need to establish a performance-based Flexible Load Fast Track. While the DC Flexibility Joint Filers support the overall thrust of these comments, endorsement does not necessarily extend to every individual statement or characterization.

² *Interconnection of Large Loads to the Interstate Transmission System*, Docket No. RM26-4-000 (Oct. 27, 2025).

deliver flexibility to the grid. The FLFT would be an expedited (60-90 day) interconnection process for loads that commit to flexibility and would incorporate this flexibility into studies that determine whether transmission grid upgrades are necessary to interconnect the load. Recognizing flexibility as a measurable, technology-neutral, and performance-based capability will accelerate deployment of data-center and industrial facilities, enhance reliability, and keep electricity costs affordable by deferring or eliminating the need for new, expensive, long-lead time infrastructure. This rulemaking offers an opportunity to align U.S. energy reliability with national competitiveness and technological leadership.

III. BACKGROUND AND POLICY CONTEXT

The Secretary of Energy’s directive of October 23, 2025, calls for prompt Commission action to ensure that large loads such as AI data centers and advanced manufacturing facilities can interconnect to the transmission system efficiently and on equal terms.

Historically, the Commission’s efforts to standardize generator interconnections through Orders 888, 2003, and 2023 created transparency and certainty for new supply resources. A comparable framework for large loads is now needed. Current state-by-state processes are inconsistent, opaque, and time-consuming, resulting in multi-year delays that slow economic growth amid an AI revolution that presents a critical opportunity for U.S. innovation, competitiveness, and prosperity.

The United States is witnessing the largest single-decade increase in industrial electricity demand in over forty years.³ Data centers serving AI and digital services often require hundreds of megawatts to more than a gigawatt of generation capacity to power them, comparable to traditional generation projects.⁴ Standardized interconnection procedures that recognize the flexible operating characteristics will allow these facilities to connect more quickly and at lower overall system cost.

IV. AEMA’S OVERARCHING POSITION

AEMA supports the Commission’s proposal to standardize interconnection procedures for large loads and stridently recommends that the final rule:

1. Create a Flexible Load Fast Track (FLFT) for facilities that make verifiable flexibility commitments, with clear timelines for studies and approvals within the FLFT.
2. Direct transmission owners and Regional Transmission Operators (“RTOs”) to provide transparency to large loads on the specific constraints and resulting grid upgrades, if any, their grid connection would cause, and the necessary actions to address the constraints and defer or avoid the upgrade.
3. Define flexibility using performance-based criteria that are technology-neutral and encompass software and hardware approaches.

³ U.S. Energy Information Administration (EIA), *Electric Power Monthly: Table 5.1—Sales of Electricity to Ultimate Customers by End-Use Sector*, October 2025.

⁴ U.S. Department of Energy, Office of Policy, *The Electricity Demand of Artificial Intelligence and Data Centers*, October 2025.

4. Recognize secure, standardized operator–facility interfaces that allow operators to request and verify curtailment while facilities retain operational control.
5. Coordinate with regional entities and NERC to foster consistent implementation and reliability oversight without prescribing specific technologies.

These actions will reduce administrative burden, improve planning accuracy, and enable the acceleration of data-center and industrial growth while ensuring system stability and affordability.

V. FLEXIBILITY AS A GRID RESOURCE

AEMA generally supports each of the principles outlined in the ANOPR. Most importantly, AEMA enthusiastically supports the ANOPR’s Principle 7, which calls for expedited interconnection studies for large loads that agree to be curtailable or dispatchable. Flexibility is a grid resource that should be rewarded through faster study and approval. The 60-day timeline proffered by the ANOPR is reasonable and nearly aligns with the 90-day that the Commission approved for Southwest Power Pool as part of their “High Impact Large Load” Policy.

The Commission, with its issuance and implementation of Order Nos. 719 and 745 is well aware of the resilience, reliability and economic benefits of load flexibility. A flexible facility that can modulate a portion of its demand during system peaks reduces the need for grid infrastructure, creating value for that facility and all other customers in the market. The Commission should direct transmission providers to treat this verifiable reduction as equivalent to a system-impact mitigation, allowing such projects to be studied and interconnected on a shorter timeline. The FLFT rewards verifiable outcomes rather than prescribing any particular asset or control architecture.

- I. AEMA recommends that the Commission direct transmission owners and RTOs to establish a dedicated Flexible Load Fast Track process, parallel to the existing serial queue. The FLFT would allow facilities meeting defined flexibility thresholds to receive interconnection study approval within 60 to 90 days. For the FLFT, the Commission should also expedite the processes for “Interconnection Agreements” for Large Loads and the construction of physical upgrades. After studies are completed, these agreements and construction can sometimes take years to complete. Such a timeline directly contradicts the ANOPR’s urgency and goal for “efficient, timely, and non-discriminatory load interconnections.”

Critically, this FLFT commitment should be defined and bounded. The flexibility offered by a large load is a specific, quantifiable resource, not an open-ended obligation. AEMA recommends that the interconnection study process be used to define clear, upper-bound limits on the required flexibility (e.g., maximum number of hours per year, hours per event, depth of flexibility, and events per year) needed to mitigate specific system constraints. This approach is based on established reliability modeling and ensures that data center operators have the operational certainty required to invest in these high-value facilities, achieving both grid needs and economic productivity.

To comply with Principle 7 of the ANOPR and for large loads to decide whether they can be curtailable, transmission owners and RTOs must provide transparency during the study process regarding the specific constraints that a large load would create and the frequency and amount of curtailment that would be necessary. Most existing interconnection study processes lack this transparency.

Figure 1 illustrates the transparency that Transmission Owners and RTOs should provide to large load customers during the interconnection process. In this example, the large load must curtail its grid usage by 100 MW from 5 AM-9AM and 6PM-8PM in January and February and by 150 MW from 4PM-9PM in June, July, and August to defer or avoid the need for new infrastructure. If customers can curtail during these hours, they can have an expedited interconnection process.

Required MW Reduction												
Hour ending	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1												
2												
3												
4												
5												
6	-100	-100										
7	-100	-100										
8	-100	-100										
9	-100	-100										
10												
11												
12												
13												
14												
15												
16												
17						-150	-150	-150	-150			
18	-100	-100				-150	-150	-150	-150			
19	-100	-100				-150	-150	-150	-150			
20						-150	-150	-150	-150			
21												
22												
23												
24												

II. DEFINING FLEXIBILITY

The final rule should define flexibility in functional terms that encourage innovation. AEMA recommends four categories:

1. Temporal flexibility: The ability to shift or modulate computing or industrial power consumption over time in response to operator signals or price events.
2. Spatial flexibility: The capability to shift computing or industrial activity geographically to relieve local congestion or reliability constraints.
3. Resource flexibility: The coordinated use of on-site assets such as storage, thermal systems, or other controllable resources to manage total facility load.

4. Adjacent flexibility: The ability to coordinate with nearby storage, aggregated third-party resources, or other controllable resources so that the combined effect reduces net withdrawals on the locally constrained parts of the transmission system.

Software-based controls, hardware-based assets (such as storage or on-site generation), and hybrid approaches are all valid methods for delivering these forms of flexibility. The final rule should remain strictly technology-neutral, focusing not on the *method* used, but on the *measurable, verifiable, and reliable performance* of the resources at their point of interconnection.

III. SYSTEM-OPERATOR CONTROL AND VERIFICATION

System reliability requires that operators have confidence in the performance of flexible loads. This can be achieved through a secure, standardized operator–facility interface—which may be implemented through software, hardware controls, or both—rather than direct physical utility control of on-site systems.

Under this model, the operator issues a dispatch signal, for example a 100-megawatt curtailment request to the facility's registered energy management interface. The facility's flexibility management platform executes the reduction automatically and provides real-time or near-real-time, auditable telemetry back to the operator to confirm compliance. This approach mirrors long-standing successful demand-response verification practices, maintains cybersecurity, and respects operational autonomy while giving operators dependable visibility into performance.

IV. RELIABILITY AND COST BENEFITS

Integrating flexibility into the interconnection process improves reliability and keeps energy costs in check. Fast-acting load adjustments can reduce the frequency of local congestion events, defer transmission upgrades, and moderate wholesale price volatility.

For example, a 500-megawatt data-center facility that can curtail 100 megawatts during system peaks effectively provides a reliability service comparable to new peaking generation. The avoided cost of network reinforcement and generation capacity benefits all consumers. For high-value computing, curtailment should be targeted and infrequent, tied to defined system conditions and used only as necessary to protect reliability. For hardware-based assets (such as storage or on-site generation), flexibility should be dispatched according to the operating characteristics of the resources. The FLFT balances fast access to power with predictable operating conditions, making flexibility a practical option for economically critical workloads.

Flexibility also allows data centers and manufacturers to meet their growth potential while enabling the power system to maintain cost discipline. AI infrastructure is essential to national security, productivity and innovation; incorporating flexibility ensures that this growth enhances, rather than strains, the grid.

V. ALIGNMENT WITH OTHER ANOPR PRINCIPLES

AEMA's recommendation for a performance-based Flexible Load Fast Track (FLFT) can be a key mechanism for implementing several of the Commission's other principles in a cohesive, market-based manner.

- Principle 3 (Study Coordination): The FLFT provides a framework for studying co-located or adjacent generation and load together. By studying the "net" withdrawal or injection, as this principle suggests, the FLFT can accurately value the reduced system impact, leading to deferred upgrades and shorter study timelines.
- Principle 4 (Readiness Requirements): AEMA supports standardized readiness requirements. The FLFT can incorporate clear, transparent criteria (such as deposits and site control) consistent with generator interconnection, appropriately scaled for large loads to deter speculative projects.
- Principle 5 (Hybrid Facilities): The FLFT can be the pathway to appropriately study hybrid facilities. A data center that commits 100 MW of verifiable flexibility (potentially spanning on-site storage, workload orchestration, and adjacent distributed energy resources) should be studied based on its "net" firm withdrawal rights, consistent with this principle.
- Principle 6 (System Protection): The secure, verifiable interface described in Section VI can be the modern "system protection facility" called for in Principle 6. It is the software-based protection that provides auditable assurance that the load will curtail when required within its defined operating limits, thereby preventing unauthorized withdrawals and protecting the system.
- Principle 8 (Cost Allocation): AEMA believes the FLFT provides a crucial, market-based choice for large loads: (1) interconnect as an inflexible load and potentially bear the costs for the resulting, often slow and expensive, network upgrades; or (2) interconnect as a flexible load via the FLFT, avoid triggering those upgrades, and get online faster at lower cost. This protects all other ratepayers from socializing upgrade costs.
- Principles 11 & 12 (Transmission and Ancillary Service): To operationalize the FLFT, AEMA supports new transmission service products. A "Non-Firm Withdrawal Right," for example, could be an appropriate tariff-based underpinning. This clarification ensures facilities can combine firm service for mission-critical loads with a separate, curtailable portion, which would in turn have different ancillary service responsibilities (Principle 12).
- Principles 13 & 14 (Transition and NERC Coordination): AEMA supports a clear transition plan (Principle 13) that allows projects already in the queue to opt in to the FLFT if they meet its flexibility and readiness criteria. We also support coordination with NERC (Principle 14) to establish uniform telemetry and performance reporting standards, which can enable the "verifiable" aspect of the FLFT.

VI. IMPLEMENTATION AND TRANSITION

To ensure effective implementation, AEMA recommends that the Commission:

1. Direct transmission providers to file standardized Flexible Load Fast Track procedures within 12 months of the final rule.

2. Permit existing queue participants to opt in to the FLFT process if they can demonstrate verified flexibility.
3. Coordinate with NERC to establish uniform telemetry and measurement standards.
4. Hold a technical conference within six months of final rule issuance to share best practices and develop a common reporting framework across regions.

In the interim, ISOs should pilot flexible-load interconnection processes and report results within one year. These steps will accelerate learning, promote uniformity, and prevent regional disparities.

VII. CONCLUSION

AEMA appreciates the Commission's leadership in addressing the challenges posed by rapid growth in large industrial and data-center loads. Standardizing interconnection procedures and recognizing flexibility as a key reliability asset will enable continued U.S. leadership in AI and manufacturing while maintaining affordable and reliable power for all consumers.

AEMA respectfully urges the Commission to adopt a final rule that establishes a Flexible Load Fast Track, defines flexibility broadly in a technology-neutral and performance-based manner, and ensures that large-scale load growth contributes to both reliability and national competitiveness. We look forward to working with the Commission as it moves through this rulemaking process.

Respectfully submitted,
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On behalf of the Advanced Energy Management Alliance
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